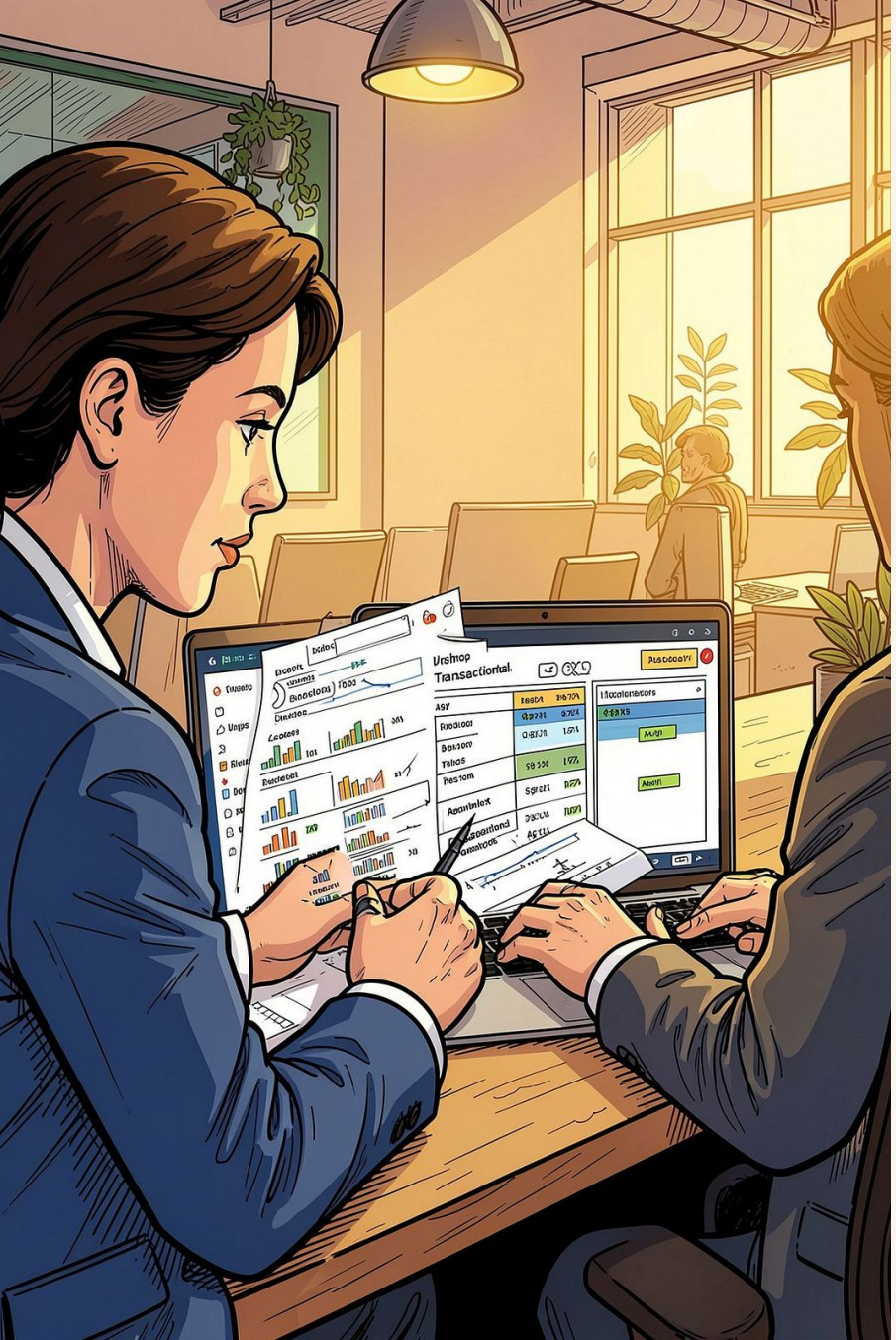


Customer Shopping Behavior Analysis

Analysis of 3,900 transactions to reveal spending patterns, segments, product preferences, and subscription behavior to guide strategy.

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Dataset at a Glance

Rows & Columns

3,900 rows · 18 columns

Key Features

Demographics, purchase details, behavior, subscription, shipping

Missing Data

37 missing values in Review Rating

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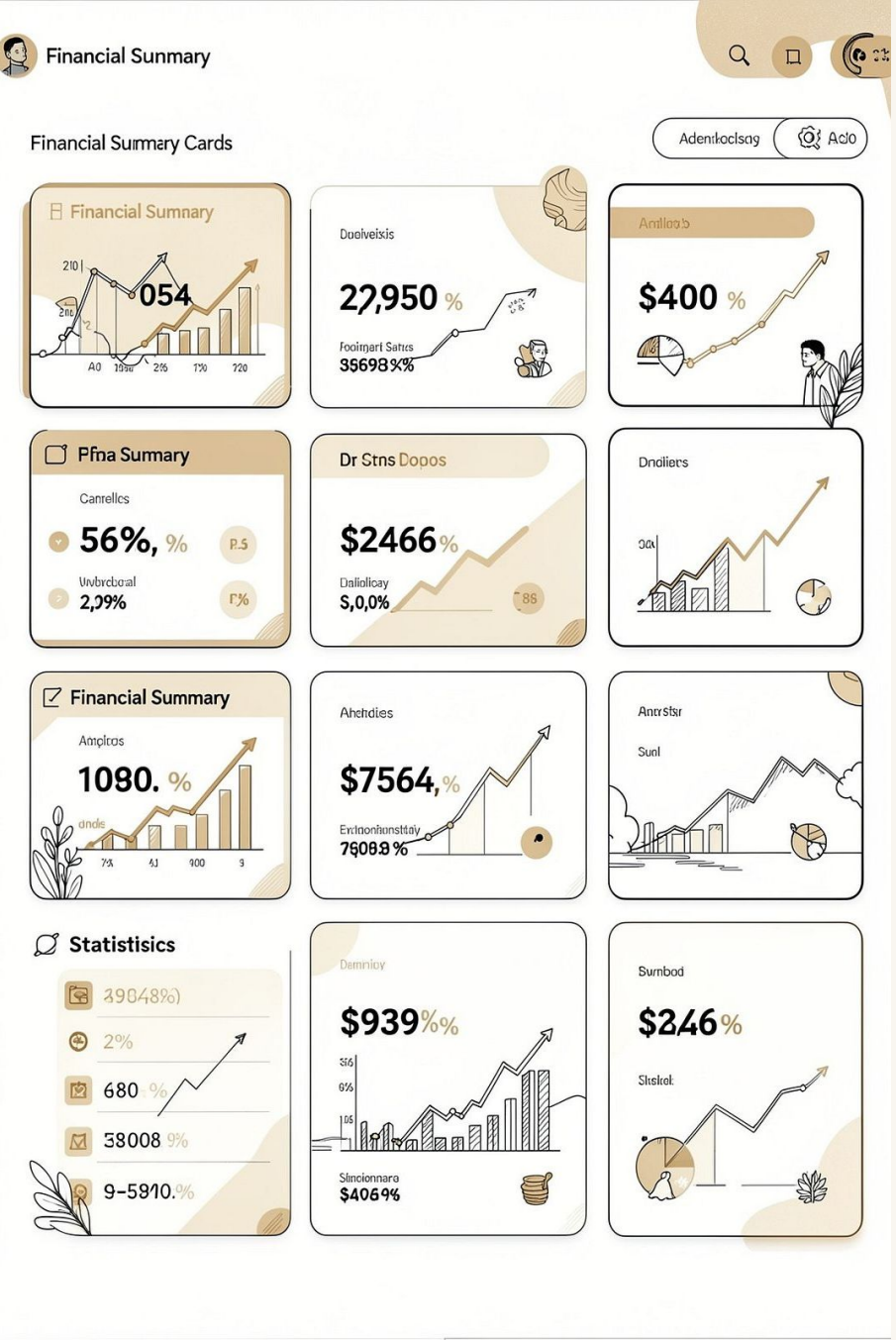
Data Preparation (Python)

Loaded with pandas; used df.info() and .describe() for structure and summary. Standardized column names to snake_case. Imputed missing review ratings with category medians.

Engineered age_group and purchase_frequency_days. Dropped redundant promo_code_used. Integrated cleaned data to PostgreSQL.



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Descriptive Stats Snapshot

Customer Age

Mean 44.07 • Range 18–70 • 25% = 31, 75% = 57

Purchase Amount

Mean $59.76 • Min $20 • Max $100 • Std $23.69

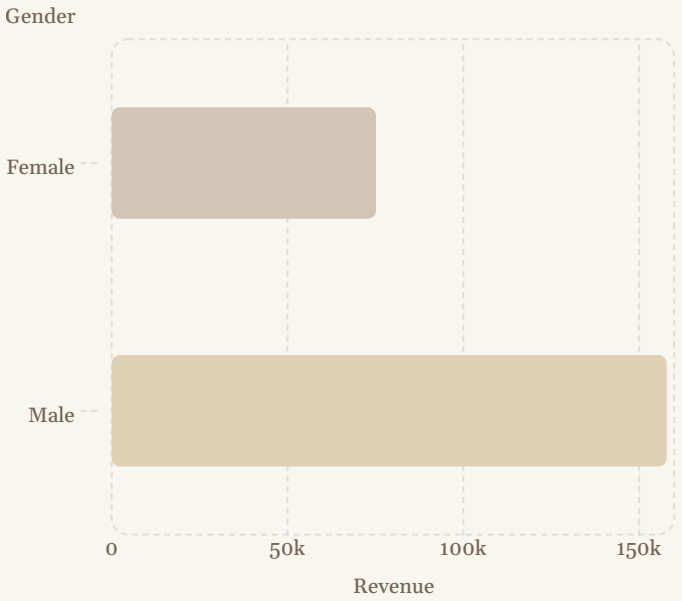
Review Rating

Mean 3.75 (37 values imputed by category median)

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Revenue by Gender (SQL)

Male customers generated higher total revenue in the dataset. Use this to refine targeting and product assortment strategies.





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Top Products & Ratings



Gloves

Avg Rating 3.86



Sandals

Avg Rating 3.84



Boots

Avg Rating 3.82

Also high: Hat (3.80) and Skirt (3.78). Highlight top-rated items in campaigns and product pages to boost conversion.

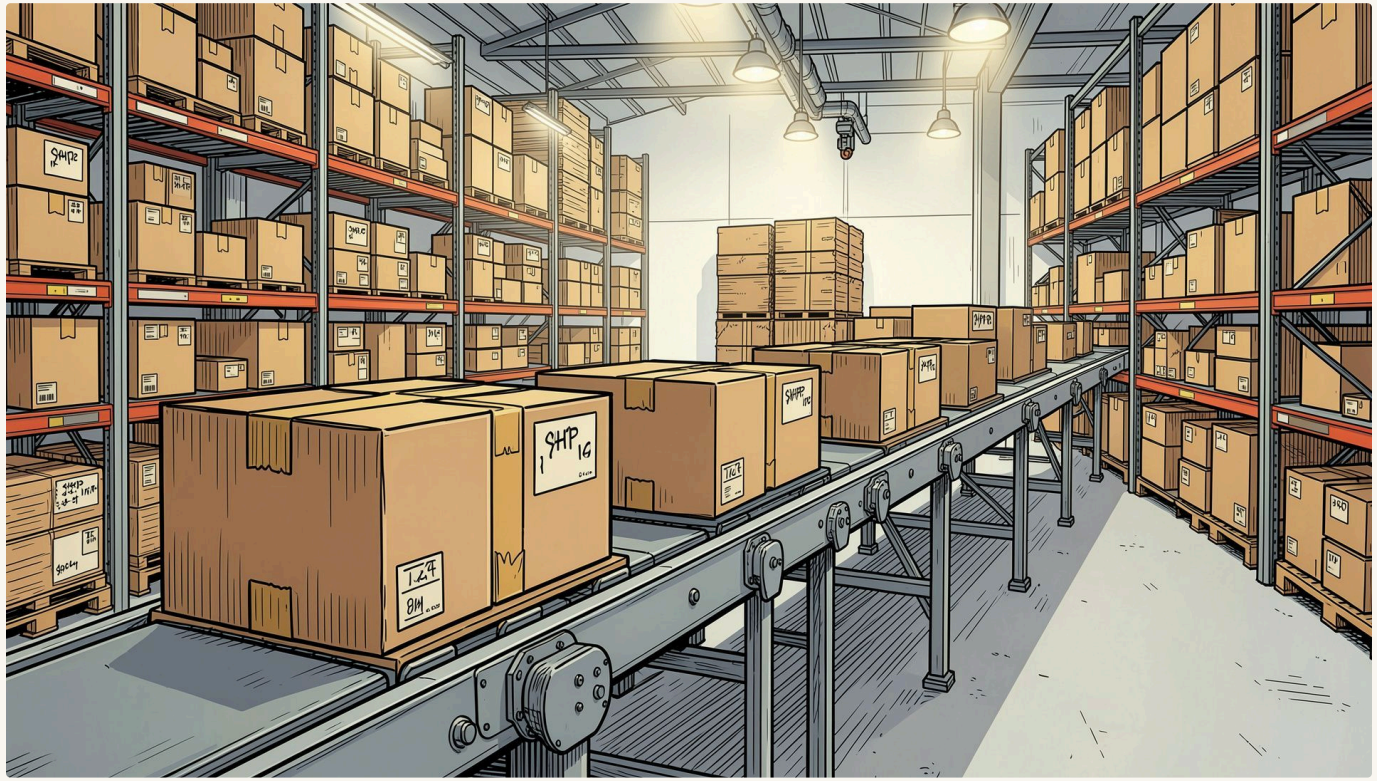


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Shipping & Purchase Behavior

Average purchase: Standard $58.46 vs Express $60.48. Subscriptions: 1,053 subscribers avg spend $59.49 (total $62,645); Non-subscribers 2,847 avg $59.87 (total $170,436).

Repeat buyers (>5 purchases): 958. Loyalty segment dominates (3,116 loyal customers).



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Discounts & High-Spenders

Identified discount users who still spent above average—839 rows flagged. Top discount-dependent items: Hat (50.00%), Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), Pants (47.37%).

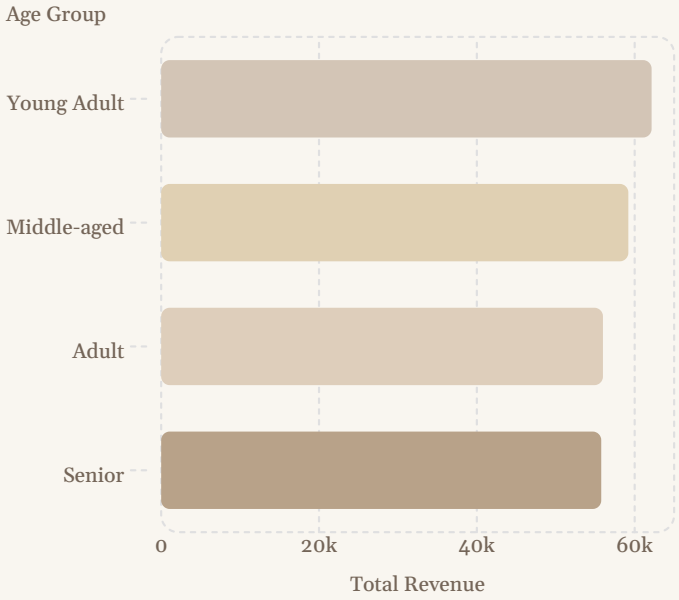
Recommendation: review discount policy to balance conversion and margins.



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Revenue & Segmentation Insights

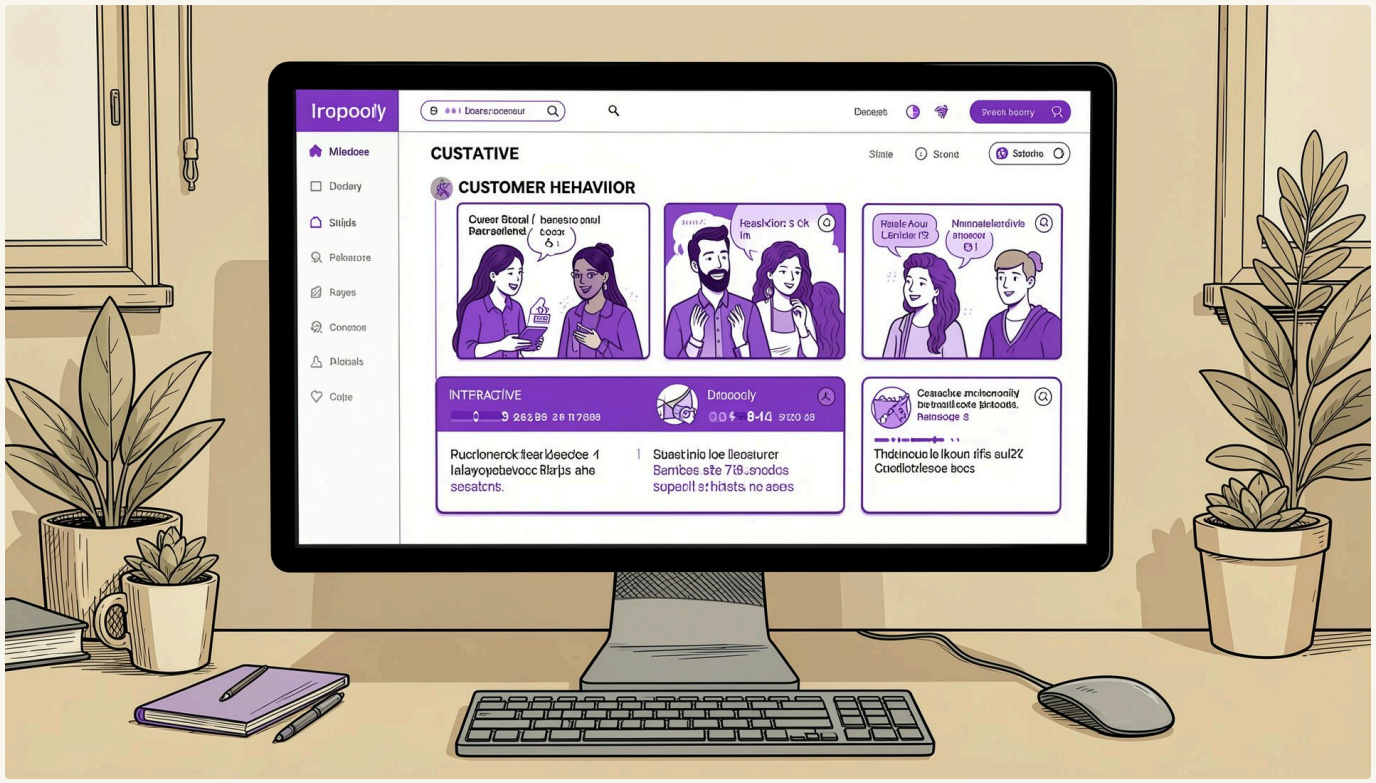
Young Adults contribute the largest revenue. Customer segments: Loyal 3,116; Returning 701; New 83. Prioritize targeted marketing for high-revenue age groups and loyalty programs for repeat buyers.





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Dashboard & Business Recommendations



Boost Subscriptions

Promote exclusive benefits to increase subscriber base.

Loyalty Programs

Reward repeat buyers to move more customers into Loyal segment.

Review Discounts

Balance discount-driven sales with margin controls.

Product Positioning

Feature top-rated and best-selling items in campaigns.

Targeted Marketing

Focus on high-revenue age groups and express-shipping users.

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